

4.6.4

EE25BTECH11016 - Taraka Abhinav

Question:

Find the equation of a line passing through the point $(2, 3, 2)$ and parallel to the line $\mathbf{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$. Also, find the distance between these two lines. (12, 2018)

Solution:

Part 1: Equation of the Parallel Line

This remains the same regardless of the method for finding the distance.

$$\mathbf{a}_2 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (1)$$

$$\Rightarrow \text{Line Equation is } \boxed{\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}} \quad (2)$$

Part 2: Distance using Least Squares Method

The distance is the magnitude of the error vector when projecting the connecting vector $\mathbf{w} = \mathbf{a}_2 - \mathbf{a}_1$ onto the direction vector $\mathbf{b}$. The vectors are:

$$\mathbf{a}_1 = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{w} = \mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \quad (3)$$

The least squares solution for the projection scalar t is found from the normal equation:

$$(\mathbf{b}^T \mathbf{b})t = \mathbf{b}^T \mathbf{w} \quad (4)$$

Calculating the dot products:

$$\mathbf{b}^T \mathbf{b} = \|\mathbf{b}\|^2 = (2)^2 + (-3)^2 + (6)^2 = 49 \quad (5)$$

$$\mathbf{b}^T \mathbf{w} = (2)(4) + (-3)(0) + (6)(2) = 20 \quad (6)$$

Solving for t :

$$49t = 20 \Rightarrow t = \frac{20}{49} \quad (7)$$

The shortest distance vector is the error vector, $\mathbf{d} = \mathbf{w} - t\mathbf{b}$.

$$\mathbf{d} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} - \frac{20}{49} \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 - 40/49 \\ 60/49 \\ 2 - 120/49 \end{pmatrix} = \frac{1}{49} \begin{pmatrix} 156 \\ 60 \\ -22 \end{pmatrix} \quad (8)$$

The distance is the magnitude (norm) of this error vector:

$$d = \|\mathbf{d}\| = \frac{1}{49} \sqrt{156^2 + 60^2 + (-22)^2} = \frac{\sqrt{28420}}{49} = \frac{14 \sqrt{145}}{49} \quad (9)$$

$$\Rightarrow d = \boxed{\frac{2 \sqrt{145}}{7}} \quad (10)$$

